DataCo Supply Chain — SQL Analytics Report

Dataset: DataCo Smart Supply Chain for Big Data Analysis (Kaggle)
Scope: 180,000+ order-item records across global markets
Tools: PostgreSQL, custom normalized schema across 6 tables
Files: 09 SQL files covering schema design through advanced analysis

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1. Executive Summary

The DataCo supply chain operation generates **$14.5M in revenue** across 79,000 fulfilled orders but operates at a thin **12.3% overall margin** with nearly **1 in 5 orders (18.7%) being loss-making**. The most critical operational finding is that **over half of all shipments (~55%) are delivered late** — a rate that is strikingly uniform across all shipping modes, markets, order quantities, days of the week, and months of the year. This consistency rules out any targeted cause and points to a systemic scheduling methodology problem: delivery windows are set too optimistically across the board. First Class shipping is the sharpest symptom of this — it carries a near-100% late delivery rate in every region and every month, not because of carrier failure but because its scheduled window is structurally impossible to meet. Revenue is geographically concentrated in Western Europe and Central America, but both carry below-average margins compared to smaller regions like Southern Africa and Central Asia. The discount strategy is actively eroding profitability: 94% of orders carry a discount, the most popular discount tier (11–20%) produces the lowest average profit, and mid-value first-time customers generate the strongest long-term returns — suggesting acquisition spend is being directed at the wrong customer type.

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2. Data Overview

Schema Design

The raw flat file was normalized into six tables to enable clean, join-based analysis:

Table	Grain	Primary Key
`orders`	One row per order line item	`order_item_id`
`customers`	One row per unique customer	`customer_id`
`product_details`	One row per product	`product_card_id`

Table	Grain	Primary Key
`category_details`	One row per category	`category_id`
`shipping_details`	One row per shipment	`order_item_id`
`sales_details`	One row per order item	`order_item_id`

Key Cleaning Decisions

- `customer_email` dropped — fully masked (XXXXX) across all rows, no analytical value
- `product_description` dropped — NULL for all rows in source data
- `order_date_date` and `shipping_date_date` split into separate DATE and TIME columns
- `late_delivery_risk` cast from INTEGER to BOOLEAN; `product_status` same
- Two analysis views created: `orders_clean` (COMPLETE/CLOSED status only, positive quantities) for financial analysis; `orders_active` (excludes CANCELED/SUSPECTED_FRAUD) for volume analysis
- Derived columns added: `profit_flag` (profitable/breakeven/loss_making), `shipping_delay_days` (actual minus scheduled), `delivery_performance` (early/on_time/late)

Validation Results

- No null values found in any join or aggregation columns
- No duplicate `order_item_id` values
- Zero orphaned foreign key references across all tables
- `benefit_per_order` and `order_profit_per_order` fully consistent (0 contradictions)

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3. Findings

3.1 Revenue and Profitability

Overall business health is positive but margin is thin.
The business fulfils ~79,000 orders generating $14.5M revenue and $1.78M profit at a 12.3% overall margin and $183 average order value. While revenue scale is healthy, the slim margin leaves limited buffer for operational inefficiencies — particularly the widespread late delivery problem documented in section 3.4.
SQL technique: multi-metric aggregation with `NULLIF` guard on margin calculation

18.7% of all orders are loss-making.
Nearly 1 in 5 orders destroys value rather than creating it. These are not outliers — they represent a structural pattern concentrated in heavily discounted orders and specific category-region-mode combinations identified in section 3.7.
SQL technique: `profit_flag` *derived column using* `CASE WHEN`*, grouped aggregation across flag categories*

Top revenue categories diverge from top margin categories.
Fishing, Cleats, Camping & Hiking, Cardio Equipment and Women's Apparel lead on profit. These same categories also appear in the profit leakage analysis (section 3.7), meaning their profitability is fragile when combined with the wrong shipping mode or region — particularly when routed through First Class or into high-late-rate areas.
SQL technique: three-table join (orders → product_details → category_details) with `margin_pct` *as derived column*

Top revenue products and top volume products are not the same.
Field & Stream Sportsman 16 Gun Fire Safe leads by revenue but ranks seventh by volume. Perfect Fitness Perfect Rip Deck leads by volume but does not appear in the revenue top 10. This split reveals two distinct product roles — high-ticket low-frequency items driving revenue and low-ticket high-frequency items driving volume — which should be managed with separate pricing and stocking strategies.
SQL technique: two separate `ORDER BY` *queries on the same aggregation —* `total_revenue DESC` *vs* `total_units DESC`

3.2 Customer Segments

All three segments are commercially similar — segment alone does not predict customer value.
Consumer leads in total revenue and profit, followed by Corporate and Home Office. However the differences are not dramatic: average order values range from $983 to $1,009 and margins are near-identical across all three. Segment is not a strong standalone predictor of profitability.
SQL technique: join across orders_clean and customers, grouped aggregation with margin calculation

Repeat purchase rate is consistently ~70% across all segments.
Corporate leads marginally at 70.4%, Home Office at 70.0%, Consumer at 69.5%. The near-identical rate across segments suggests retention mechanisms (or lack thereof) affect all customers equally — no segment is being retained meaningfully better than others. A 30% single-purchase rate across the board represents a significant revenue recovery opportunity.
SQL technique: CTE computing per-customer order count, outer query using `SUM(CASE WHEN order_count > 1)` *for repeat rate*

Corporate customers deliver the highest CLV and longest engagement lifespan.
Corporate avg CLV: $1,009 over 240 days active. Consumer: $992 over 235 days. Home Office: $983 over 238 days. Home Office profit per customer ($115.6) is notably lower than Corporate ($124.7) and Consumer ($123.4) despite a similar lifespan — suggesting Home Office orders skew toward discounted or lower-margin products.
SQL technique: CTE with `MAX(order_date) - MIN(order_date)` *for lifespan, outer aggregation by segment*

3.3 Discount Strategy

94% of all orders carry a discount — the business is over-reliant on discounting.
Across every month of the dataset, 94–95% of orders include a discount at some level. Average discount rate is stable at approximately 10.1–10.3% throughout the full date range with no seasonal pattern. Discounting is not a targeted promotional tool here — it is effectively the default pricing mode.
SQL technique: monthly aggregation of `pct_discounted` *and* `avg_discount_rate` *from orders_clean*

The most popular discount tier is also the lowest-profit tier.
The High (11–20%) discount bucket contains the most orders (30,856) but produces the lowest average profit ($20.46) and highest loss rate (19%). No-discount orders produce the highest average profit ($26.51) with a comparable loss rate (18.2%). If discounts were driving incremental volume that offsets margin loss, the no-discount loss rate would be higher — the data does not support this.
SQL technique: `CASE WHEN` *bucketing on* `order_item_discount_rate`, `loss_rate_pct` *as* `SUM(CASE WHEN profit < 0)/COUNT(*)`

Discount rate is not drifting upward over time — but it is not declining either.
Rolling 3-month discount rate stays flat at approximately 10.1–10.3% across 2015–2018. Rolling margin oscillates between 11% and 13% with no clear directional trend. The discount strategy is locked in: neither improving nor worsening, but consistently preventing the margin expansion the business would achieve by reducing discount frequency.
SQL technique: rolling 3-month `AVG()` *window function over monthly discount rate and margin, ordered by year and month*

3.4 Operational Performance — Delivery

Late delivery is systemic, not targeted — ~55% late rate across every cut of the data.
Late delivery rate is 54–58% across all shipping modes, all markets, all regions, all order quantities, all days of the week, and all months of the year. This uniformity is the most important operational finding in the dataset. A mode-specific or region-specific problem would have a targeted fix. A uniform problem across every dimension points to a single root cause: the scheduled delivery windows used at order

creation are systematically too optimistic.
SQL technique: `SUM(CASE WHEN delivery_status = 'Late delivery' THEN 1 ELSE 0 END)/COUNT(*)` *grouped independently by mode, market, region, quantity, DOW, and month*

First Class is structurally broken — near-100% late rate in every region, every month.
First Class shows 100% late delivery rate across all 23 regions and all 12 months without exception. Its actual average shipping duration is ~2 days, but its scheduled window is set tighter than that, making on-time delivery mathematically impossible. This is not a carrier problem — it is a contract/configuration problem. Correcting the scheduled window for First Class alone would reclassify a large portion of "late" shipments to "on time" with zero operational change.
SQL technique: late rate by shipping_mode × order_region and by shipping_mode × month, using consistent `SUM/COUNT` *pattern*

Delay magnitude is low even when late — worst case is 4 days.
Among late orders, average delay when late is 0.56–0.57 days across all markets. Worst-case delays are 4 days (Second Class), 2 days (Standard Class), 1 day (First Class and Same Day). Orders are not catastrophically delayed — they are consistently missing tight windows by small margins. This reinforces the scheduling-window diagnosis: the issue is the contract, not the carrier's execution.
SQL technique: `AVG(CASE WHEN shipping_delay_days > 0 THEN shipping_delay_days END)` *isolating delay magnitude among late orders only*

Shipping mode mix is nearly identical across all regions.
Every region — from Western Europe to Central Africa to Canada — uses Standard Class for approximately 54–67% of orders, with Second Class, First Class and Same Day making up the remainder in consistent proportions. There is no region that meaningfully varies its mode mix. This is relevant because it means mode reassignment (e.g., shifting away from First Class in high-late-rate regions) cannot be the solution — every region already uses the same mix.
SQL technique: `COUNT(*) * 100.0 / SUM(COUNT(*)) OVER (PARTITION BY order_region)` *for within-region mode share*

Lead time is stable — no degradation as volume grows.
Average actual shipping days oscillate between 3.45 and 3.57 across every month from 2015 to 2018, with month-on-month changes never exceeding ±0.08 days. Transit performance is not worsening as order volume scales, which rules out capacity strain as a contributor to late delivery.
SQL technique: monthly `AVG(days_for_shipping_real)` *with* `LAG()` *for month-on-month change*

3.5 Geographic Patterns

Revenue is concentrated in Western Europe and Central America, but neither leads on margin.
The top three revenue combinations are Western Europe/EE.UU. ($1.45M), Central America/EE.UU.

($1.37M) and Western Europe/Puerto Rico ($876K). However Western Europe carries a 12.79% margin and Central America 12.06% — both below the margins achieved by smaller regions. The business's highest-revenue markets are not its most profitable ones.
SQL technique: three-level geographic GROUP BY (market → region → country) with margin_pct derived column

Southern Africa and Central Asia are the highest-margin regions despite low volume.
Southern Africa: 15.91% margin. Central Asia: 15.62%. These are the two highest-margin regions in the dataset, both well above the 12.3% overall average. West Asia (10.48%) and Eastern Europe (10.57%) are the lowest-margin regions. The business is unknowingly over-investing operationally in low-margin high-volume regions while under-investing in high-margin regions.
SQL technique: grouped aggregation by order_region and market with `SUM(profit)/SUM(revenue)` *margin formula*

High late-rate regions are driven entirely by First Class and Second Class — not by geography.
South of USA (61.4%), South Asia (60.0%) and Central Asia (59.7%) lead on late delivery rate. But the region × mode cross-analysis shows that within every region, First Class is 100% late and Second Class is 74–91% late — while Standard Class is 36–52% late and Same Day is 20–73% late. The regional variation in overall late rate is entirely explained by variation in mode mix within those regions, not by any structural geographic difficulty.
SQL technique: two-dimensional GROUP BY (order_region × shipping_mode) with `SUM/COUNT` *late rate*

3.6 Time and Seasonality

Order volume peaks in January, May, July–August and is lower in Q4 — not the typical retail pattern.
January leads with 7,819 orders, followed by August (7,043) and July (6,870). November and December are among the lowest months (5,573 and 5,568). This is an inverted retail seasonality — the dataset likely reflects B2B or sports/outdoor purchasing cycles rather than consumer holiday behaviour.
SQL technique: `EXTRACT(MONTH FROM order_date)` *with* `TO_CHAR` *for readable month names, joined to shipping_details for late rate*

Late delivery rate does not spike with order volume — confirming the systemic diagnosis.
Despite January being the highest-volume month, its late rate (55.97%) is actually the lowest of any month. August has one of the highest late rates (58.61%) despite average volume. There is no correlation between order volume and late delivery rate across any month or day of the week, which rules out surge capacity as a driver.
SQL technique: monthly and DOW aggregation of both order count and late rate in the same query

Day-of-week has minimal effect on order volume or late delivery rate.
Total orders range from 11,129 (Saturday) to 11,458 (Sunday) — effectively flat across the week. Late delivery rate ranges from 55.91% (Saturday) to 58.20% (Tuesday) — a 2.3 percentage point spread with

no meaningful pattern. Orders and deliveries are distributed evenly across the week.
SQL technique: `EXTRACT(DOW FROM order_date)` *with* `TO_CHAR` *for day names*

Revenue is effectively flat year-over-year — and 2018 data is truncated.
2015 and 2016 show near-identical revenue (~$4.86M each). 2017 shows a marginal decline (-5.4%). The 2018 figure (-97%) is an artefact of the dataset ending mid-year (January only), not a real business decline. Rolling 3-month revenue stays in a tight band of $385K–$452K throughout the full period, confirming the business is stable but not growing.
SQL technique: yearly aggregation with `LAG()` *for prior-year revenue,* `revenue_growth_pct` *as* `(current - prior)/prior * 100`

September and October show the highest average profit per order.
September averages $26.48 profit per order (total: $184.7K), October $28.43 ($159.4K). January and March are the lowest at $21.58 and $19.32. The discount rate is virtually identical across all months (~10.1–10.3%), so the profit variation is not driven by seasonal discounting — it likely reflects category mix (more high-margin products ordered in autumn).
SQL technique: monthly aggregation of `avg_discount_rate`, `pct_discounted`, `avg_profit` *and* `total_profit` *from orders_clean*

3.7 Advanced Analysis

Composite risk scoring identifies Second Class × Fishing/Cardio/Camping combos as highest risk.
Combining late rate, average delay magnitude and order volume into a composite NTILE score, the highest-risk combinations (score 9/9) are all Second Class shipments of Fishing, Cardio Equipment and Camping & Hiking into Caribbean, Southeast Asia and Eastern Asia. These represent high-frequency, high-delay combinations in moderate-to-high revenue categories — the most operationally exposed part of the business.
SQL technique: three-CTE structure: raw aggregation → `NTILE(3)` *scoring across three dimensions → composite score summation, with* `HAVING COUNT(*) >= 50` *to exclude noise*

Customer retention drops sharply after month 1 but stabilises — classic e-commerce decay curve.
All cohorts show 100% retention at month 0 (first purchase). Month 1 retention drops to approximately 8–21% depending on cohort. Retention then stabilises in the 10–16% range through months 2–23 before tapering to near-zero. Early cohorts (Jan–Nov 2015) show longer retention tails, likely because more time has elapsed for repeat purchases. Late 2017 cohorts show truncated tails due to dataset end date.
SQL technique: four-CTE cohort structure using `DATE_TRUNC('month')` *and* `AGE()` *for month-difference calculation, avoiding the year-boundary error of simple* `EXTRACT(MONTH)` *subtraction*

Mid-value first-time customers generate the strongest long-term returns.
Customers whose first order was $50–$200 (mid bucket) go on to place an average of 6.33 subsequent orders and generate $1,086 in lifetime revenue — higher than customers who started with a high first

order (>$200), who average 4.73 subsequent orders and $1,020 lifetime revenue. Low first-order customers (<$50) lag significantly at $593 lifetime revenue. The implication is that acquisition spend optimised for large first orders is not the best predictor of long-term value — mid-range first orders identify the most engaged customers.
SQL technique: `DISTINCT ON (order_customer_id)` ordered by `order_date ASC` to isolate first orders, joined to lifetime aggregation CTE

Profit leakage is concentrated in Standard Class and First Class routes for Water Sports, Men's Footwear and Indoor/Outdoor Games.
The leakage analysis (high volume quartile + lowest margin quartile) surfaces 17 specific combinations. The most frequent patterns are Water Sports and Men's Footwear via Standard Class into Pacific Asia and USCA, and Fishing/Camping/Cardio via First Class into LATAM. Average discount rates on these leakage combinations range from 9–11%, confirming that moderate discounting on already-thin-margin categories in high-late-rate routes is the primary value destruction mechanism.
SQL technique: dual `NTILE(4)` ranking on volume and margin independently, filtered to `volume_quartile = 1 AND margin_quartile = 1`

The single highest-priority intervention is Western Europe / Fishing — $242K revenue at risk.
The operational recommendation query scores region × category combinations by combined late rate, revenue at risk and volume. Western Europe / Fishing leads with 1,134 orders, a 59.5% late rate and $242K revenue at risk (priority score 7). Western Europe / Cleats follows with 1,588 orders, 60.0% late rate and $156K at risk. At priority score 8, West Asia / Fishing and Caribbean / Cardio Equipment rank highest but at lower absolute revenue exposure. The Western Europe combinations represent the best trade-off between scale and intervention impact.
SQL technique: three-CTE structure aggregating late rate + revenue at risk + volume, then `NTILE(3)` scoring each dimension and summing into priority_score

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4. Limitations

No true processing time data. `shipping_date` equals `order_date` across all records, meaning the order-to-ship gap cannot be measured. Processing bottlenecks cannot be distinguished from transit bottlenecks with this dataset.

`late_delivery_risk` is a precomputed flag. The dataset includes a pre-built binary risk flag rather than deriving lateness from raw timestamps. All analysis used `delivery_status` as the primary source of truth and treated the flag as supplementary.

Customer geography is approximate. Customer city and country come from registration data, not delivery address — shipping regions from the orders table may not always match the customer's actual location.

No returns or refund data. Loss-making orders are identified through negative profit values but the root cause (returns, fraud, excessive discounting) cannot be distinguished from available columns alone.

2018 data is truncated. The dataset ends in January 2018, making year-over-year comparisons for 2018 misleading. All trend analysis treated 2018 figures as partial-year only.

Product descriptions fully absent. All `product_description` values are NULL, limiting any text-based product characterisation or search-intent analysis.

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5. Recommendations

1. Fix First Class scheduled delivery windows immediately — zero operational cost.
First Class shows 100% late delivery rate across every region and every month without exception. Actual shipping duration averages ~2 days but the scheduled window is set tighter. Adjusting the contracted window to reflect actual carrier performance would reclassify the majority of First Class "late" shipments to "on time" at no cost. This is the highest-leverage, fastest fix in the dataset.
Supported by: file 05 Section 2, file 07 Section 5, file 08 Section 7

2. Audit the High (11–20%) discount tier, starting with Water Sports and Men's Footwear.
This tier contains the most orders but the lowest average profit, and these specific categories appear in the profit leakage analysis when routed via Standard Class into Pacific Asia and USCA. A targeted review of whether these discounts are driving incremental orders or subsidising purchases that would happen anyway could recover meaningful margin. The stable 10.1% rolling discount rate across all years shows this is structural, not promotional — it needs a deliberate policy change.
Supported by: file 06 Section 5, file 09 Section 3

3. Prioritise Western Europe / Fishing and Western Europe / Cleats for operational intervention.
These two combinations have the highest absolute revenue at risk ($242K and $156K respectively) with late rates above 59%. Given that the late rate problem is systemic rather than geographic, the intervention here should be mode-level: move these high-value category flows away from First Class and Second Class and into Standard Class where late rates are 36–42% rather than 80–100%. This is the only mode shift that can materially reduce late delivery exposure without changing carrier relationships.
Supported by: file 09 Section 7, file 07 Section 2 and Section 5

4. Refocus customer acquisition on mid-value first-order customers.
Customers with a $50–$200 first order generate higher lifetime revenue ($1,086) and place more subsequent orders (6.33) than high first-order customers ($1,020, 4.73 orders). Retention drops sharply after month 1 across all cohorts, suggesting the acquisition-to-retention handoff is the primary leakage point. Improving month-1 re-engagement (personalised follow-up, mid-range product recommendations) could disproportionately improve CLV given that early cohort retention tails extend

to 20+ months.

Supported by: file 06 Sections 2 and 4, file 09 Sections 2 and 4